Homework 5: Logic and Circuits (15 points)

Due to Gradescope by 11:45 PM on Tuesday, September 27th
You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade

Homework Goal: Practice evaluating Boolean expressions, writing truth tables, and following/creating circuits.

Vocab

1. Practice your vocab from these chapters! (1 point)
   We moved away from using vacuum tubes, and now use transistors, devices that can be ON or OFF, and have no mechanical/moving parts. Transistors can be combined to create gates, which in turn can be combined to create circuits, which transform a set of binary inputs into a set of binary outputs.
   Tautology is a word used to describe a Boolean expression that is never false.

Truth Tables

2. Provide a truth table for each of the following types of gates: (2 points)
   The rows must be ordered correctly.
   a. 2-input NAND gate
   b. 2-input NOR gate
   c. 3-input AND gate
   d. 3-input OR gate

NAND

A	B	A NAND B
0	0	1
0	1	1
1	0	1
1	1	0

NOR

A	B	A NOR B
0	0	1
0	1	0
1	0	0
1	1	0

AND

A	B	C	ABC
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

OR

A	B	C	A+B+C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

3. Fill in this truth table for the following Boolean expression: (2 points)
((a NOR c) AND (NOT b)) OR c

a	b	c	out
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

Boolean Logic

4. Assume that a = 1, b = 4, and c = 3. What is the result of each of the following Boolean expressions? (2.5 points)
   a. (a > 1) OR (b == c)

   F or F

   (F)

   b. ((a + b) > c) AND (b < c)

   4 < 3

   T and F

   (F)

   c. NOT (a == 1)

   (F)

   d. NOT ((a == b) OR (b == c))

   (F or F)

   (T)

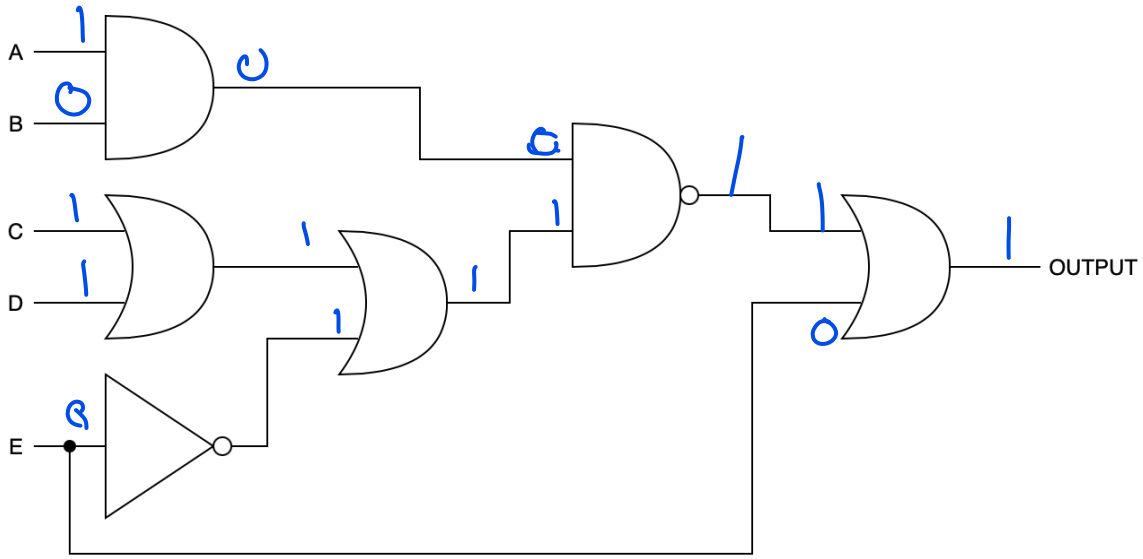
   e. (a == 1) AND (b == 1) AND (c == 2)

   T F F

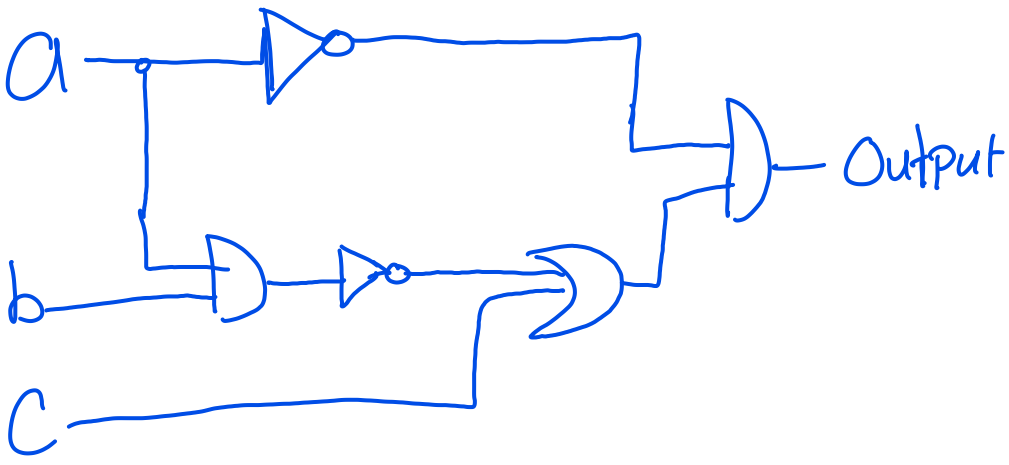
   (F)

Circuits

5. Given the following circuit diagram, what is the value output when the inputs are
   A = 1, B = 0, C = 1, D = 1, E = 0 (don't miss the two circles in the circuit)? (1.5 points)



6. Draw the circuit diagram for the following Boolean expression (2 points)
   (NOT a) AND (c OR (NOT (b AND a))



7. Given the following truth table, write a corresponding Boolean expression and then draw the circuit. (4 points)

a	b	c	out
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

(NOT A) and B and C
A and NOT(B) and NOT(C)

(NOT A) AND B AND C + A AND (NOT B) AND (NOT C)

Output = A'BC + AB'C'